Module 2

State-Variable-Based Modeling and Simulation

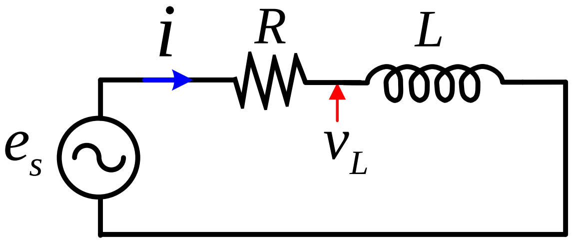
Most Important Topics and Concepts

❑ Principles of:

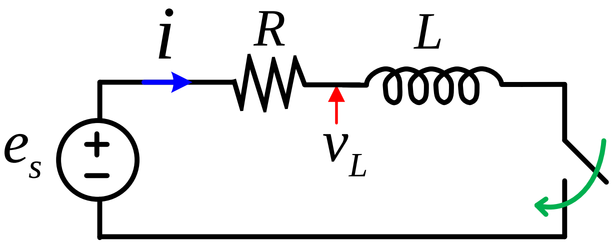
- State-space modeling of electric circuits
- Integration methods and ODE solvers
- Numerical stability and stiffness
- Time step size control
- Multi-step methods

Electric Circuits Solution

❖ Phasor Analysis:



❖ Transient Analysis:

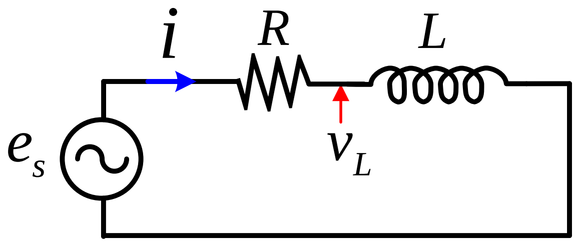


❖ Analytical Solution:

❖ Numerical Solution:

Electric Circuits Solution

❖ ODE vs. DAE:



- **Ordinary Differential Equations (ODEs):** $\dot{\mathbf{x}}(t) = \mathbf{f}\left(\mathbf{x}(t),\ \mathbf{u}(t),\ t\right)$
  - ✓ All equations are differential in nature (i.e., involve derivatives of states)
- **Differential-Algebraic Equations (DAEs):** $\mathbf{F}\left(\dot{\mathbf{x}}(t),\ \mathbf{x}(t),\ \mathbf{z}(t),\ \mathbf{u}(t),\ t\right) = 0$
  - ✓ Some equations do not have derivatives (i.e., are algebraic constraints)

✓ **ODEs:** Suitable for control systems and small circuits with well defined states (requires selecting independent states and eliminating algebraic constraints

✓ **DAEs:** Scales more easily and systematically to larger networks but is harder to solve due to handling of constraints

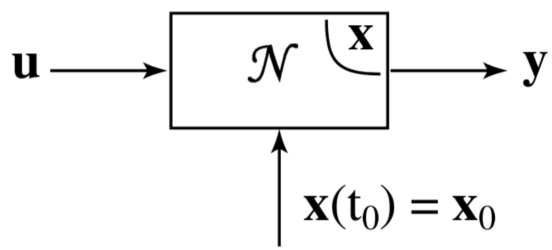
State-Variable-Based Modeling

- Many physical systems (electrical, mechanical, thermal, etc.) are described by **differential equations**

✓ State-space modeling provides a **systematic way** to represent these equations in a compact vector form

❖ State-Space Form:

✓ A **Linear Time-Invariant (LTI)** system can be expressed as:



$$\begin{cases} \dfrac{d\mathbf{x}(t)}{dt} = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \\ \mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t) \end{cases}$$

$\mathbf{x}:$

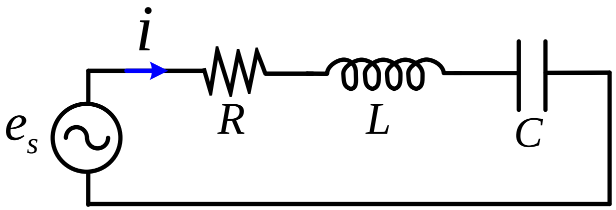
$\mathbf{u}:$

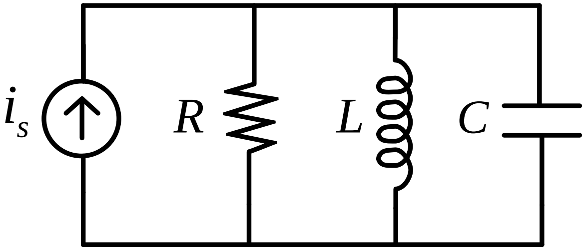
$\mathbf{y}:$

✓ **State variables** are the minimum set of variables describing the system's memory (e.g., capacitor voltages, inductor currents, etc.)

State-Variable-Based Modeling

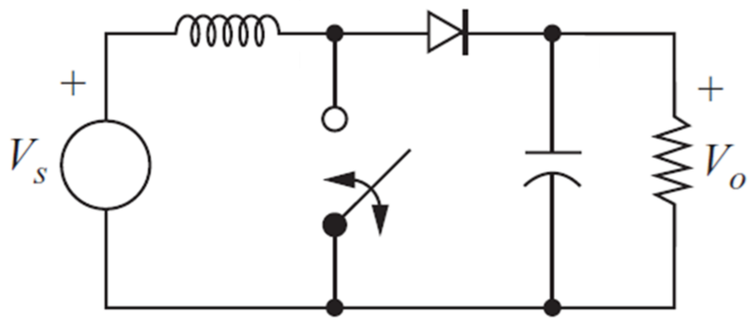
❖ **Example 1:** Express the equations of the circuits below in state-space form:





State-Variable-Based Modeling

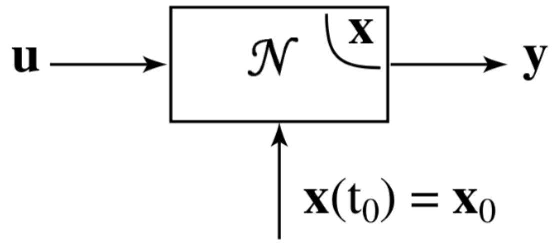
❖ **Example 2:** Express the equations of the converter below in state-space form:



State-Variable-Based Modeling

- Many real systems are **nonlinear** and **cannot** be expressed in a matrix form

✓ We can write the equations in a general form that can also be applicable to nonlinear systems



❑ General State-Space Form:

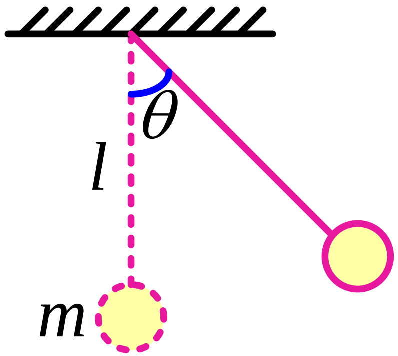
$$\begin{cases} \dfrac{d\mathbf{x}(t)}{dt} = \mathbf{f}\left(\mathbf{x}(t),\ \mathbf{u}(t),\ t\right) \\ \mathbf{y}(t) = \mathbf{g}\left(\mathbf{x}(t),\ \mathbf{u}(t),\ t\right) \end{cases}$$

$\mathbf{f},\ \mathbf{g}:$

❖ Examples of Real Nonlinear Systems:

State-Variable-Based Modeling

❖ **Example 3:** Express the equations of the pendulum below in state-space form:



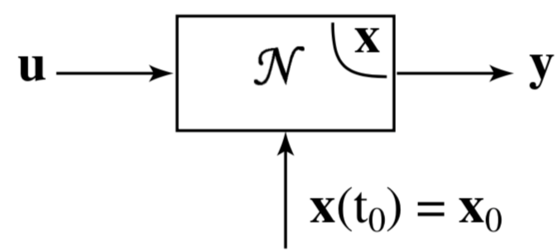
State-Space Models and Linearization

- Nonlinear systems are **hard to analyze** directly but can be converted into an **LTI** system around an operating point

✓ Then we can obtain transfer functions for controller tuning, small-signal stability analysis, predict transient behavior (via poles/zeros), etc.

❑ General State-Space Form:

$$\boxed{\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, \mathbf{u}, t) \quad , \quad \mathbf{y} = \mathbf{g}(\mathbf{x}, \mathbf{u}, t)}$$



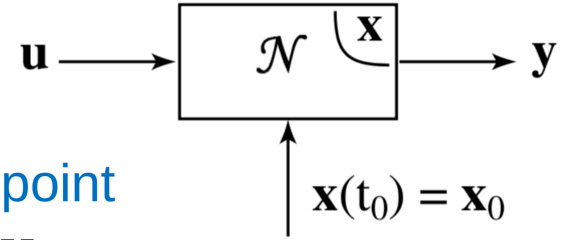
- Assume operating around $\mathbf{x}_{\mathrm{op}} \;,\; \mathbf{u}_{\mathrm{op}}$
- Apply perturbation to the states and inputs

✓ observe the changes in the dynamics and outputs

$$\begin{cases} \mathbf{x} = \mathbf{x}_{\mathrm{op}} + \Delta\mathbf{x} \\ \mathbf{u} = \mathbf{u}_{\mathrm{op}} + \Delta\mathbf{u} \end{cases} \Rightarrow \boxed{\begin{cases} \dfrac{d\Delta\mathbf{x}}{dt} = \mathbf{A}\Delta\mathbf{x} + \mathbf{B}\Delta\mathbf{u} \\ \Delta\mathbf{y} = \mathbf{C}\Delta\mathbf{x} + \mathbf{D}\Delta\mathbf{u} \end{cases}}$$

State-Space Models and Linearization

❖ Linearization of Nonlinear State-Space System:



$$\boxed{\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x},\mathbf{u},t) \quad , \quad \mathbf{y} = \mathbf{g}(\mathbf{x},\mathbf{u},t)}$$

operating point

$$\mathbf{x}_{\mathrm{op}} \quad , \quad \mathbf{u}_{\mathrm{op}}$$

$$\begin{cases} \mathbf{x} = \mathbf{x}_{\mathrm{op}} + \Delta\mathbf{x} \\ \mathbf{u} = \mathbf{u}_{\mathrm{op}} + \Delta\mathbf{u} \end{cases}$$

$$\begin{cases} \dfrac{d\mathbf{x}}{dt} = ? \\ \mathbf{y} = ? \end{cases}$$

Use Taylor Series!

Transfer Function in Laplace Domain

- ✓ Linear systems can be transformed into the **Laplace domain**

$$\begin{cases} \dfrac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \\ \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u} \end{cases}$$

- ✓ We can obtain the input-output transfer function:

$$\mathbf{H}(s) = \frac{\mathbf{y}(s)}{\mathbf{u}(s)}$$

$$\mathbf{H}(s) = \mathbf{C}\left[s\mathbf{I} - \mathbf{A}\right]^{-1}\mathbf{B} + \mathbf{D}$$

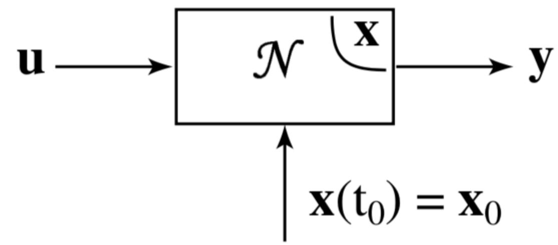
State-Variable-Based Modeling

State-Space of LTI System

$$\begin{cases} \dfrac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \\ \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u} \end{cases}$$

General State-Space Form

$$\begin{cases} \dfrac{d\mathbf{x}}{dt} = \mathbf{f}\left(\mathbf{x},\ \mathbf{u},\ t\right) \\ \mathbf{y} = \mathbf{g}\left(\mathbf{x},\ \mathbf{u},\ t\right) \end{cases}$$



❑ Strictly Proper System:

When the output is not directly related to input (i.e., **no algebraic feedthrough**)

linear system:

nonlinear system:

❑ Algebraic Loop:

When a variable is defined implicitly by itself through algebraic (non-differential) relations, i.e., $z = g\left(\mathbf{x},\ z,\ t\right)$

you must solve the algebraic equation first (often iteratively)

can slow down simulations!

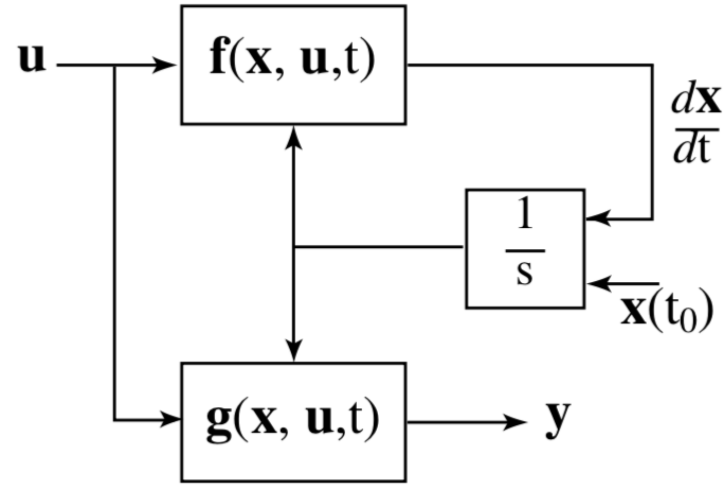
❖ **Example:**

State-Variable Solution

- **Objective:** To find a **numerical solution** for the state-space equation below:

General State-Space Form

$$\begin{cases} \dfrac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x},\ \mathbf{u},\ t) \\ \mathbf{y} = \mathbf{g}(\mathbf{x},\ \mathbf{u},\ t) \end{cases}$$



✓ Assume there is **no algebraic loop**

✓ Assume input $\mathbf{u}$ is part of the definition of the function $\mathbf{f}$ ⇒ treat $\mathbf{u}$ as a **known function of time** and incorporate it inside $\mathbf{f}$

Standard Canonical Form

$$\begin{cases} \dfrac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, t) \\ \mathbf{y} = \mathbf{g}(\mathbf{x}, t) \end{cases} \qquad \mathbf{x}(t_0) = \mathbf{x}_0$$

⇒ ✓ The numerical algorithms used to solve this set of ODEs are called **ODE SOLVERS**

State-Variable Solution

To be continued…